

Lab 02 – MATH 240 – Computational Statistics

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Abstract

In this lab we aimed to take data from over 100 json files and two csv files, clean the data, combine what we wanted to keep from each set of data, and eventually, use our final data frame to find out which artist contributed more in the song *Allentown*.

Keywords: data frames;

1 Introduction

The purpose of this lab was to extract data from json files and store them in a data frame, extract data from a csv file and store them as a data frame, and then finally to clean up, organize, and export data from data frames into csv files.

2 Methods

The data we used in this lab came from 181 json files, *EssentiaModelOutput.csv*, and *LIWCOutput.csv*

2.1 Extract Data from a json File

For this first part we practiced creating an object for a json file using `fromJSON` from the `jsonlite` package (Ooms, 2014) and extracting the track, album, and artist from the name of the file, and extracting data from within the json file. To do so, I created an object for the name of the file (*The Front Bottoms-Talon Of The Hawk-Au Revoir (Adios).json*), then I used `str_split_i()` from the “stringr” package (Wickham, 2023) to split the name of the example json file at the dashes, and isolate the artist, album, and track. For the track, however, since it comes last in the name has the “.json” at the end and I needed to use `str_sub()` to remove it.

Next, I created an object for the file path of the json by using `paste()` to combine the string “EssentiaOutput/” with the name of the file so I could extract overall loudness, spectral energy, dissonance, pitch salience, tempo in beats per minute, beat loudness, danceability, and tuning frequency.

2.2 Repeat for every File

Now, to repeat this for every file so that I could store the music data in a data frame, I created a vector with all the files in the *EssentiaOutput* folder using `list.files()`. Then I used a `for` loop to extract overall loudness, spectral energy, dissonance, pitch salience, tempo in beats per minute, beat loudness, danceability, and tuning frequency from each json.

This data contributes to all of the music data collected on the selected songs and will help us to determine which artist contributed more on “Allentown,” so I used `cbind()` to combine all of the vectors for the artist, album, track and values that I had just extracted from the json files into a data frame.

2.3 Cleaning Data from *EssentiaModelOutput.csv*

The first step to this process is using `read.csv()` create a data frame with information from the spreadsheet. Then our first task was to average three data sets for two values: **valence** and **arousal**. To do so, I used a `for` loop to average the three datasets into one for both **valence** and **arousal**. I then stored the newly created vectors with these averages in the *EssentiaModelOutput* data frame.

Then I repeated this process for **aggressive**, **happy**, **party**, **relaxed**, **sad**, **acoustic**, **electric**, and **instrumental**, storing all of these in the data frame.

Then, using the `names` function, I renamed `eff_timbre_bright` to `timbreBright` so that it could be more easily referenced later.

The last step for this data frame was to keep only the new columns I had just created and `timbreBright`. To do this I wrote a code so that the data frame would only save the first three columns (artist, album, track), the 48th column (`timbreBright`), and the last 10 columns (new columns I just created).

2.4 Load *LIWCOutput.csv* and Merge all Data Frames

To load *LIWCOutput.csv* I just used `read.csv` then used `merge()` to merge all of the data frames. With this new data frame I have all of the music data in one place, making it easier to compare between artists.

2.5 Coding Challenge: What does the Data Tell Us?

Using ggplot2 (Wickham, 2016) I created 3 graphs each one mapping artist on the x axis and the value for the selected column (sad, Tone, aggressive) on the y axis.

3 Results

According to the graphs, “The Front Bottoms” had a higher mean than “Manchester Orchestra” for the sad, Tone, and aggressive values. Yet, their collaboration song had the highest value out of any of the means.

4 Discussion

The evidence seems to suggest that “The Front Bottoms” contributed more to *Allentown* because the means of the values that I selected are closer to the values for *Allentown*. However, we still need to examine the data closer and more broadly to make this determination.

References

- Ooms, J. (2014). The jsonlite package: A practical and consistent mapping between json data and r objects. *arXiv:1403.2805 [stat.CO]*.
Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York.
Wickham, H. (2023). *stringr: Simple, Consistent Wrappers for Common String Operations*. R package version 1.5.1.